COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	22/07/26	**DATE SUBMITTED:**	22/07/26
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ACTIVITY NO. 1.1 **Algorithm, Flowchart, Pseudocode**

1. PROCEDURE OUTPUT

What procedures did you perform?

Main Function

```
#include<iostream>
#include"singly_ll.h"


int main(){

    //stack
    Singlelist<char>* head = nullptr;
    Singlelist<char>* second = nullptr;
    Singlelist<char>* third = nullptr;
    Singlelist<char>* fourth = nullptr;

    //allocate memory to heap
    head = new Singlelist<char>;
    second = new Singlelist<char>;
    third = new Singlelist<char>;
    fourth = new Singlelist<char>;

    //Storing a value inside the node
    head -> data='J';
    second -> data='M';
    third -> data= 'L';
    fourth -> data = 'E';

    //linking the nodes
    head-> next = second;
    second-> next= third;
    third-> next = fourth;
    fourth-> next = nullptr;
```

```
    std::cout<<second<<std::endl;
    std::cout<<third<<std::endl;
    std::cout<<fourth<<std::endl;
    std::cout<<head->next<<std::endl;

    std::cout<<"testing of Travelsal: \n";
    ListTraversal(head);

    std::cout<<"Testing of sllinsertHead: \n";
    sllinsertHead('E',&head);
    ListTraversal(head);

    std::cout<<"Testing of sllgeneralInsert";
    sllgeneralInsert('X', head->next->next);
    ListTraversal(head);

    std::cout<<"Testing of sllinsertEnd: \n";
    sllinsertEnd('S', &head);
    ListTraversal(head);

    std::cout<<"Testing of Deleting nodes \n";
    sllDeleteNode('S', &head);
    ListTraversal(head);
    sllDeleteNode('E', &head);
    ListTraversal(head);
    sllDeleteNode('X', &head);
    ListTraversal(head);
```

```
58
59      std::cout<<"Testing of Deleting the List: \n";
60      sllDeleteList(&head);
61      ListTraversal(head);
62      return 0;
63  }
```

Singly_ll.h

```cpp
#ifndef SINGLY_LL_H
#define SINGLY_LL_H

template<typename T>
class Singlelist{
    public:
    T data;
    Singlelist<T>* next=nullptr;
};

template<typename T>
void ListTraversal(Singlelist<T>* head){

    while(head != nullptr){
        std::cout<<head->data;
        if(head->next != nullptr){
            std::cout<<"->";
        }
        head=head->next;
    }
    if (head == nullptr){
        std::cout<<"The list is empty";
        return;
    }
    std::cout<<std::endl;
}
```

```cpp
template<typename T>
void sllinsertHead(T newData, Singlelist<T>** currentHead){

    Singlelist<T>* newNode= new Singlelist<T>;

    newNode->data = newData;

    newNode->next = *currentHead;

    *currentHead = newNode;

}

template<typename T>
void sllgeneralInsert(T newData, Singlelist<T>* prevNode){
    if(prevNode == nullptr){
        std::cout<<"Previous Value cannot be NULL \n";
        return;
    }
    Singlelist<T>* newNode= new Singlelist<T>;
    newNode->data = newData;

    newNode->next = prevNode -> next;
    prevNode ->next= newNode;
}
```

```cpp
template<typename T>
void sllinsertEnd(T newData, Singlelist<T>** head){

    Singlelist<T>* newNode= new Singlelist<T>;
    Singlelist<T>* currentNode= *head;
    newNode->data = newData;
    newNode->next = nullptr;

    while(currentNode->next != nullptr){
        currentNode= currentNode->next;
    }
    currentNode->next=newNode;
}

template<typename T>
void sllDeleteNode(T findData, Singlelist<T>** head){

    if(*head == nullptr)return;

    Singlelist<T>* currNode=*head;
    Singlelist<T>* prevNode= nullptr;

    while(currNode != nullptr && currNode->data !=findData){
        prevNode= currNode;
        currNode= currNode->next;
    }
```

```
8   template<typename T>
9   void sllDeleteNode(T findData, Singlelist<T>** head){
0
1       if(*head == nullptr)return;
2
3       Singlelist<T>* currNode=*head;
4       Singlelist<T>* prevNode= nullptr;
5
6       while(currNode != nullptr && currNode->data !=findData){
7           prevNode= currNode;
8           currNode= currNode->next;
9       }
0
1
2       if(currNode==nullptr){
3           std::cout<<"The Data is not Found \n";
4           return;
5       }
6       if (prevNode==nullptr){
7           *head=currNode->next;
8       }else{
9           prevNode->next = currNode->next;
0       }
1       delete currNode;
2   }
3
```

```
94  template<typename T>
95  void sllDeleteList(Singlelist<T> **head){
96      Singlelist<T>* current = *head;
97
98      while(current != nullptr){
99          Singlelist<T>* temp= current;
100         current = current->next;
101         delete temp;
102     }
103     *head = nullptr;
104 }
105 #endif
```

Output

```
0x213269ac970
0x213269ac9b0
0x213269aca10
0x213269ac970
testing of Travelsal:
J->M->L->EThe list is emptyTesting of sllinsertHead:
E->J->M->L->EThe list is emptyTesting of sllgeneralInsertE->J->M->X->L->EThe list is emptyTesting of sllinsertEnd:
E->J->M->X->L->E->SThe list is emptyTesting of Deleting nodes
E->J->M->X->L->EThe list is emptyJ->M->X->L->EThe list is emptyJ->M->L->EThe list is emptyTesting of Deleting the List:
The list is empty
```

In this activity, we learned how to manipulate nodes of a list, insert value to nodes and how to link the nodes, how to insert at the head, end, and in the body, also how to delete nodes and the list. This activity made me realize how tricky it is to create a system that can store user data, insert, delete or manipulate that data. Overall, this activity showed me what must be mastered before one can create an efficient system that can flexibly manage user data.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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